JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

SEPTEMBER 2021 EXAMINATIONS

BIOLOGY **Time Allowed: 3 hours**

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer all questions in this section.

Use the OMR answer sheet provided to answer the questions.

Follow the instructions on the OMR sheet.

SECTION B: ESSAY QUESTIONS

Answer **FOUR** questions; **ONE** question from each course.

Ensure you read and follow all the Instructions on the cover page of the Answer Booklet.

SECTION A: MULTIPLE CHOICE QUESTIONS.

Answer ALL Questions.

1. The process of a white blood cell engulfing a bacterium is known as
   A. endocytosis.
   B. exocytosis.
   C. osmosis.
   D. diffusion.

2. Each of the following is a water plant EXCEPT
   A. Hycinth.
   B. Lilly.
   C. Cactus.
   D. Letuce.

3. In plants cells, the chloroplast are located
   A. inside the cell wall.
   B. outside the protoplasm.
   C. within the vacuole.
   D. within the cytoplasm.

4. The ability of the cockroach to live in cracks and crevices is enhanced by the possession of
   A. wings and segmented body.
   B. compound eyes.
   C. claws on the legs.
   D. dorso-ventrally flattened body.

5. The function of the scrotum is to
   A. keep the temperature of the testes equal to that of the rest of the body.
   B. keep the temperature of the testes slightly lower than that of the rest of the body.
   C. keep the temperature of the testes slightly higher than that of the rest of the body.
   D. keep the temperature of the testes at a constant 45°C.

6. RNA viruses require their own supply of certain enzymes because
   A. the viruses are rapidly destroyed by host cell defenses.

B. the enzymes translate viral RNA into proteins.
C. the enzymes cannot be made by host cells.
D. those cells do not possess RNA-RNA or RNA-DNA enzymes.

7. An infectious agent containing only protein and without nucleic acid molecule is a
A. prion.
B. toxin producer.
C. vivoid.
D. bacterium.

8. A peptide bond is formed between
A. an aldehyde group and an amino group.
B. a carboxyl group and an amino group.
C. an aldehyde group and a carboxyl group.
D. an aldehyde group and an ester group.

9. Which one of the following molecules contains amino acids?
A. Cellulose.
B. Collagen.
C. Galactose.
D. Glycogen.

10. A researcher, who wants to study the composition of a plant's sap, inserts a capillary tube into the phloem. What causes the sap to flow out of the tube?
A. Capillarity.
B. Hydrostatic pressure.
C. Root pressure.
D. Transpiration stream.

11. In terms of the number of individuals, which of the following taxa is most inclusive?
A. Order.
B. Family.
C. Class.
D. Species.

12. Unsaturated fat is in the form of
A. liquid.
B. solid.

C. gas.

D. mixture of liquid and home.

13. On the basis of susceptibility of organisms to thermal conditions, microorganisms are classified into all the following EXCEPT

A. Thermophiles.

B. Mesophiles.

C. Agarmophiles.

D. Psychrophiles.

14. RNA means

A. riboflavin nucleic acid.

B. ribonucleic acid.

C. ribosomal nucleic acid.

D. ribonucleic activity.

15. One of these is **NOT** an element in the chain of infection?

A. Infectious agent.

B. Reservoir.

C. Susceptible host.

D. Mode of defence.

16. Which of the following is a prokaryote?

A. Bacteria.

B. Virus.

C. Fungi.

D. Helminthes.

17. Who is the father of taxonomy?

A. Antonie Van Leeuwenhoek.

B. Christian Gram.

C. Robert Koch.

D. Carl Linnaeus.

18. The second scientific name of a microorganism is the name

A. scientific.

B. specific.

C. generic.
D. general.

19. Which of these organelles synthesize, refines, sorts, stores and secretes chemical products in the cell?
A. Mitochondria.
B. Lysosomes.
C. Golgi apparatus.
D. Ribosomes.

20. Cocoyam is an example of a
A. rhizomes.
B. suckers.
C. corms.
D. tendril.

21. Which of these is the worst cause of animal extinction?
A. Hunting.
B. Species introduction.
C. Habitat destruction.
D. Pesticides application.

22. The movement of molecules or ions from a region of high concentration to a region of low concentration is known as
A. photosynthesis.
B. osmosis.
C. diffusion.
D. plasmolysis.

23. The antibiotic neomycin is obtained from the bacterium
A. *Bacillus subtilis.*
B. *Penicillium notadum.*
C. *Streptomyces fradiae.*
D. *Streptomyces griseus.*

24. All EXCEPT one of the following is a characteristic of pathogenic bacterium.
A. Spread between hosts.

B. Become established in the host.

C. Repair the host immune system.

D. Obtain nutrients from the host.

25. One of these is NOT a classification of egg based on the yolk content.

A. Isolecithal.

B. Nanolecithal.

C. Mesolecithal.

D. Telolecithal.

26. Active transport is

A. the movement of molecules in and out of a cell.

B. an energy-requiring process in which ions and molecules are moved across membranes against a concentration gradient.

C. active uptake of materials.

D. use of heat to transport materials.

27. Which statement(s) are correct about glycolysis?

i. Glucose molecule is phosphorylated

ii. The 6-carbon hexose phosphate is split into three molecules of triose phosphate

iii. Triose phosphate is converted to pyruvic acid

A. i and ii

B. i and iii

C. i,ii and iii

D. ii and iii

28. Which of the following biomes is characterized by short growing seasons?

A. Deserts.

B. Tropical wet forests.

C. Arctic tundra.

D. Savanna.

29. In which of the following organisms would glycogen be stored?

A. *Spirogyra.*

B. *Chlamydomonas.*

C. *Rattus rattus.*

D. *Mangifera.*

30. The phenotype ratio resulting from Mendel's law of independent assortment is
A. 9:3:3:1
B. 3:1
C. 3:1:1
D. 6:3:3:4

31. The function of ribosome in cell is
A. protein synthesis.
B. starch synthesis.
C. transport materials.
D. lipid storage.

32. The joint between the atlas and the vertebrae allows for
A. rotary movement only
B. up and down and nodding movement only
C. rotatory and nodding movements only
D. no movement at all

33. In anaerobic respiration, glucose is converted to
A. carbon dioxide and water.
B. carbon dioxide only.
C. carbon dioxide and alcohol.
D. alcohol and water.

34. During the excretion of urea, there is also a corresponding re-absorption of water into the blood stream. This re-absorption take place in the
A. uriniferous tubules.
B. Bowman's capsule.
C. glomerulus.
D. Malpighian capsule.

35. Which of the following statement(s) is/are true of adult butterfly?
I. It is a specialist reproductive stage
II. Needs vast amounts of energy which are supplied by nectar.
III. Aids the dispersal of the species
A. II and III only
B. I and III only

C. II and III only

D. I, II and III

36. Organisms which take in complex organic molecules are called

A. Autotrophs.

B. Heterotrophs.

C. Symbionts.

D. Parasites.

37. The genotype ratio of a monohybrid cross in the F2 generation is

A. 3:1

B. 1:2:1

C. 1:1:2

D. 1:2

38. Chiasmata formation leads to

A. formation of gametes

B. creation of variability

C. formation of genes

D. formation of the different sexes

39. Techniques of sterilization were introduced by

A. Louis Pasteur

B. Robert Koch

C. Ferdinard Cohn

D. John Needham

40. Lipopolysaccharide is a major constituent in cell wall.

A. Gram-positive bacteria

B. Gram-negative bacteria

C. Fungi

D. Protozoa

41. Acquisition of naked DNA by a bacterium from its environment and incorporation in its genome is known as

A. Transformation

B. Transduction

C. Conjugation

D. Lysogenic conversion

42. HIV belongs to the family Retroviridae and subfamily

A. Lentivirinae.

B. Oncovirinae.

C. Retrovirinae.

D. None of the options.

43. Two useful substances produced by the light-dependent stage of photosynthesis are

A. ATP and ribulose

B. Ribulose and $NADP_2$

C. ATP and ethanol

D. ATP and $NADPH_2$

44. In animals cleavage produces a mass of cells called

A. Organs

B. Blastula

C. System

D. Cellular

45. Which of the following mode of nutrition is not among fungi?

A. Autotrophism.

B. Symbiosis.

C. Parasitism.

D. Saprotrophism.

46. Which one of the following provided the least supportive evidence to theory of evolution?

A. Embryology.

B. Paleontology.

C. Natural selection.

D. Geography.

47. Staining techniques are useful in microbiology because

A. stains deflect electrons propagated towards the slide

B. most microscopes are of poor quality

C. the cytoplasm of microorganisms is transparent

D. stains make the condenser and diaphragm of the microscope unnecessary

48. Microorganisms selected for probiotic use should exhibit all the following characteristics EXCEPT

A. Adhere to the intestinal mucosa of the host.

B. Be toxic and pathogenic to the host's microflora.

C. Remain viable for a long time.

D. Withstand HCl in the host's stomach.

49. Which of these statements do you consider to be CORRECT?

I. Rickettsia cause several human diseases including epidemic typhus.

II. Viruses lack the ability to grow and reproduce.

III. Viruses use the energy and protein producing machinery of cells to make new viral particles.

A. I only.

B. I & II only.

C. II & III only.

D. I, II & III.

50. Which of the following can be used in the classification of the monerans?

I. Physical appearance.

II. Nutritional requirements.

III. Cell chemistry.

A. I only.

B. I & II only.

C. II & III only.

D. I, II & III.

SECTION B: BIOLOGY ESSAY QUESTIONS

Answer FOUR questions in ALL; ONE from each Course.

BIO 001: GENERAL BIOLOGY

1. (a) What is gaseous exchange? **[1 mark]**

(b) What is respiratory surface? **[1 mark]**

(c) State **FOUR** properties required for a cell to be an effective respiratory surface. **[4 marks]**

(d) Give **FOUR** examples of respiratory surfaces **[4 marks]**

2. (a) Define the following Ecological terms: **[4 marks]**

i. Niche

ii. Microhabitat

iii. Ecosystem

iv. Population

(b) Give **TWO** reasons why a small population can easily go into extinction. **[2 marks]**

(c) Give **TWO** explanations each on how the population growth of an organism can be affected by density dependent and density independent factors. [**4 marks]**

BIO 002: BOTANY

3 (a) i.) Mention **TEN** macro-elements (excluding phosphorus) which are required by plants for healthy growth. **[2½ marks]**

ii.) State **THREE** anatomical differences between Monocot and Dicot **[3 marks]**

(b) State **THREE** importance of phosphorus to plants **[1½ marks]**

(c) State **THREE** symptoms which can be observed in a plant growing in soil that is deficient in phosphorus [**3 marks]**

4. (a) Mention **FOUR** roles of cytokinin in plant growth **[4 marks]**

(b) Define the following terms:

i. Hypogeal germination

ii. Meristem

iii. Epigeal germination

iv. Geotropism [**4 marks]**

(c) List **TWO** conditions necessary for growth. [**1 mark]**

(d) State **ONE** feature possessed by a seed or fruit dispersed by wind [**1 mark]**

BIO 003: MICROBIOLOGY

5. (a) Enumerate **TWO** major historical issues/challenges / events that contributed to the historical development of Microbiology. **[2marks]**

(b) Describe the **FOUR** major groups of fungi based on the spores produced. **[4 marks]**

(c) Give **FOUR** differences between bacteria and viruses. **[4 marks]**

6. (a) State **FOUR** differences between mutational and plasmids mediated drug resistance. **[4marks]**

(b) What is bacteriophage? **[1 mark]**

(c) Describe the life cycle of *E. Coli.* **[4 marks]**

(d) Define intermediate host. **[1 mark]**

BIO 004: ZOOLOGY

7 (a) Enumerate **FIVE** reasons why Monotremes and Marsupials are special categories of mammals. **[2½ marks]**

(b) List **FIVE** biological advancements of Amphibians over Fishes. **[2½ marks]**

(c) Explain ultrafiltration. **[5 marks]**

8 (a) Name the constituents of a balanced diet and state their importance in the diet. **[4 marks]**

(b) What are the modifications and the mechanisms associated with the following habits;

i. Filter feeding in mosquito larva **[3 marks]**

ii. Parasitic feeding in tapeworms **[3 marks]**